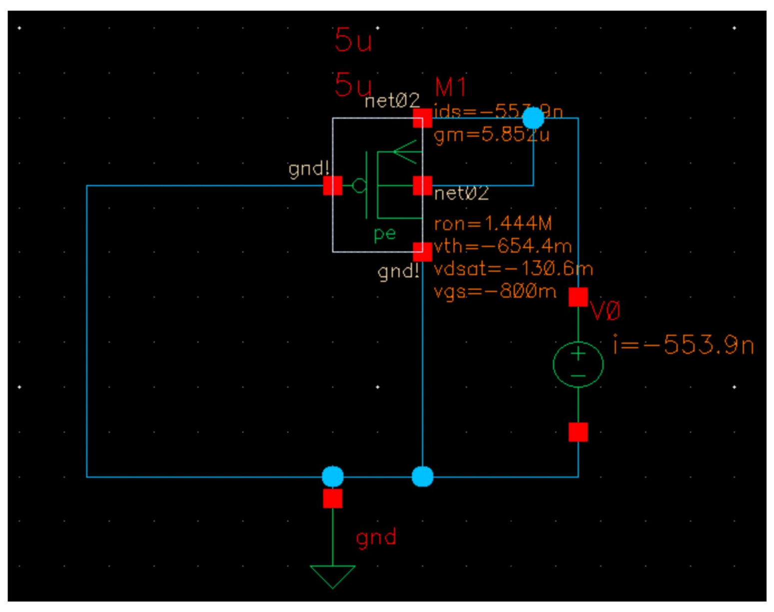
Two-Stage OpAmp Design and Transistor Characterization

34655 Integrated Analog Electronics 2

PMOS Characterization



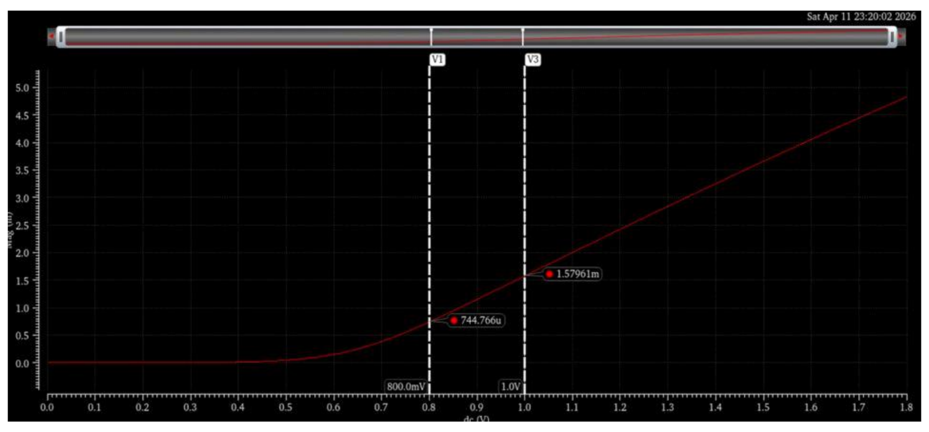
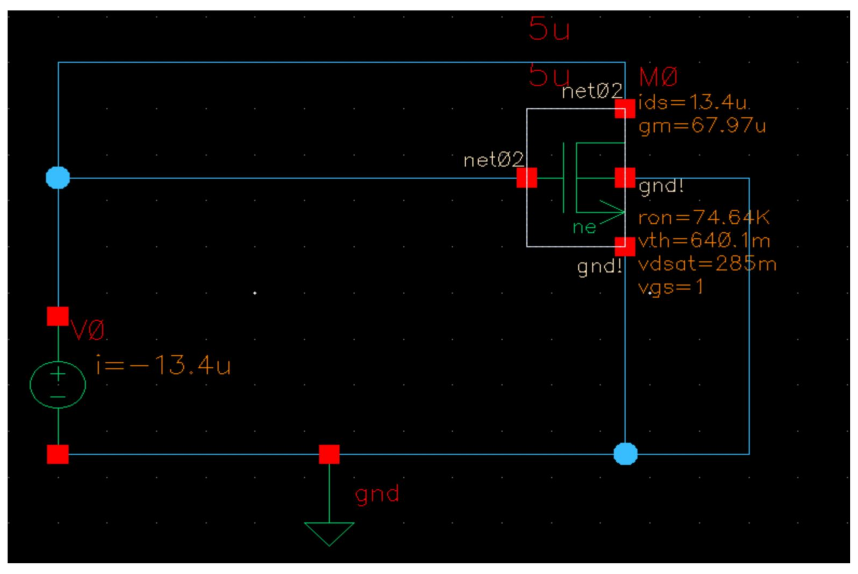


Figure 2: Shows the sqrt(I_d) against V_{GS}, given W/L=5/5

Figure 1: Circuit used to show output characteristic for PMOS

W/L [µm]	|V_GS| [V]	I_D [µA]	Slope [m√A/V]	µp·Cox [µA/V²]	|V_tp| [V]	g_m Op-point [µA/V]	g_m √(2µCox(W/L)I_D)	g_m 2I_D/V_eff [µA/V]
5/1	0.8	-2.6	8.74	36.6	0.616	29.04	28.2	28.3
5/1	1.0	-12.6	9.91	39.3	0.671	70.15	70.4	70.2
5/5	0.8	-0.6	3.82	29.2	0.597	5.825	5.92	5.91
5/5	1.0	-2.5	4.2	36.3	0.624	13.48	13.7	13.8

NMOS Characterization



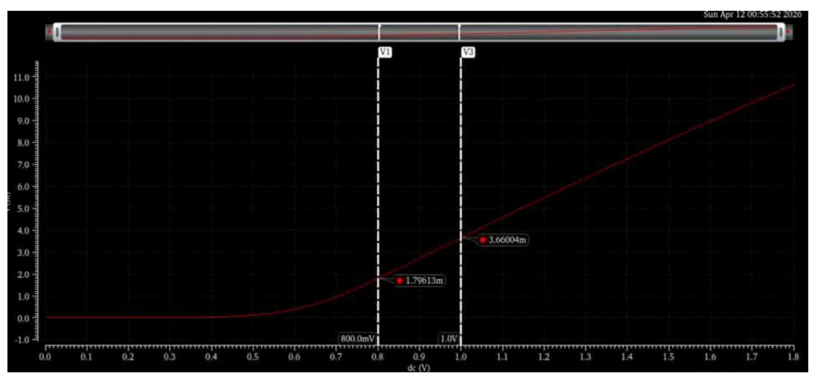


Figure 2: Shows the sqrt(I_d) against V_{GS}, given W/L=5/5

Figure 3: Circuit used to show output characteristic for NMOS

W/L [µm]	V_GS [V]	I_D [µA]	Slope [m√A/V]	µn·Cox [µA/V²]	V_tn [V]	g_m Op-point [µA/V]	g_m √(2µCox(W/L)I_D)	g_m 2I_D/V_eff [µA/V]
5/1	0.8	11.7542	19.3538	150	0.623	136.7	132.0	133.0
5/1	1.0	58.46827	21.1935	180	0.639	321.5	325.0	324.0
5/5	0.8	3.2448	8.89129	158	0.598	32.6	32.0	32.1
5/5	1.0	13.41531	9.31904	174	0.67	67.97	68.3	63.8

Requirements

The objective of this exercise was to design a two-stage, non-inverting operational amplifier, as depicted to the right. The design must also uphold the following specifications and constraints:

Midband gain	Slew Rate	Phase margin	CL bandwidth	C_A and C_B	R_1	C_L
2 V/V	>30 V/µs	>70°	>20 MHz	1 pF	1GΩ	1.5pF

The OpAmp should be modeled using the Shichman-Hodges equation for MOSFETs with channel lengths restricted to 1 µm, and the design should be limited to the 180 nm CMOS process.

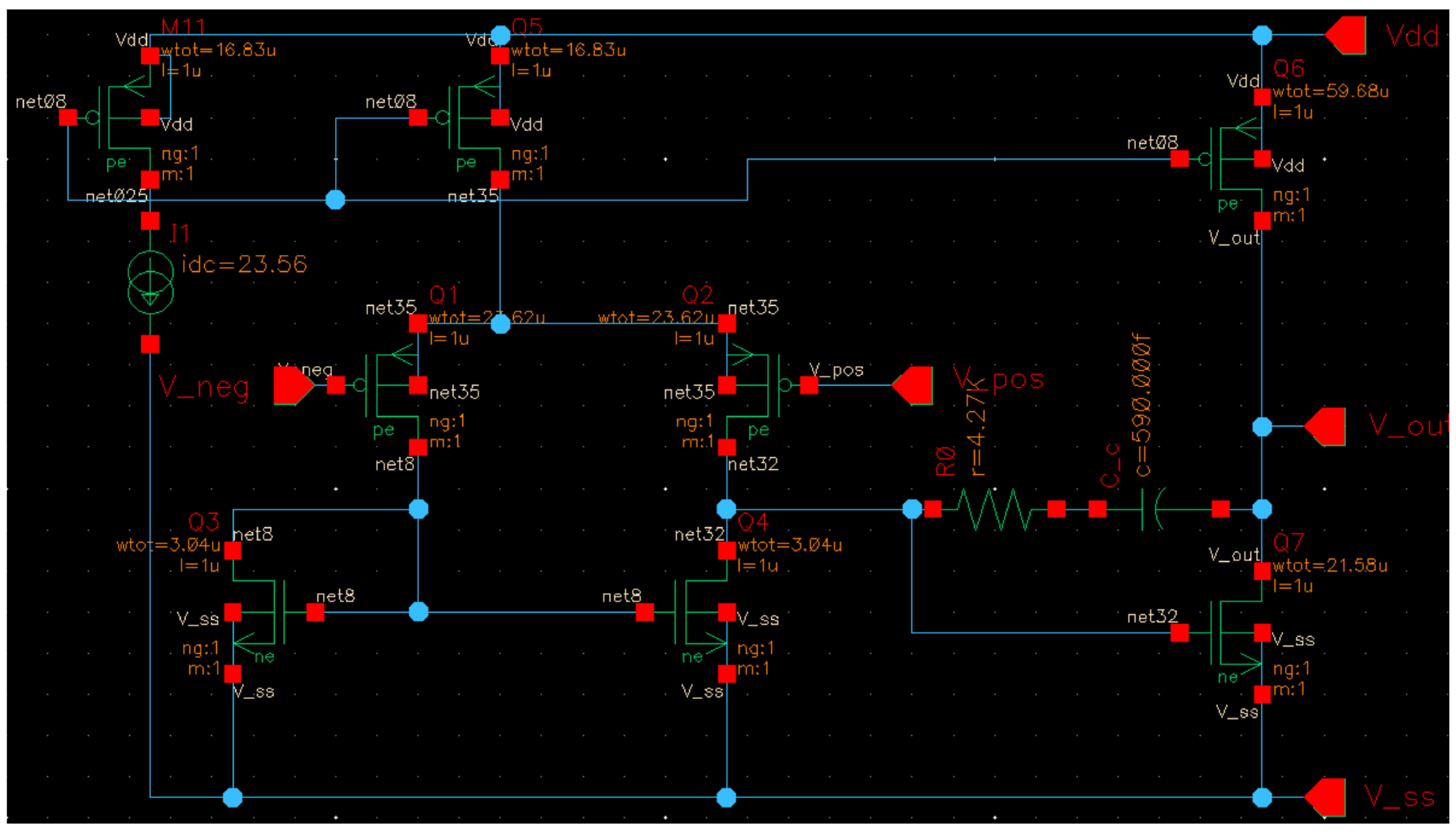


Figure 6: Shows the schematic and design parameters for the two-stage operational amplifier inside Cadence .

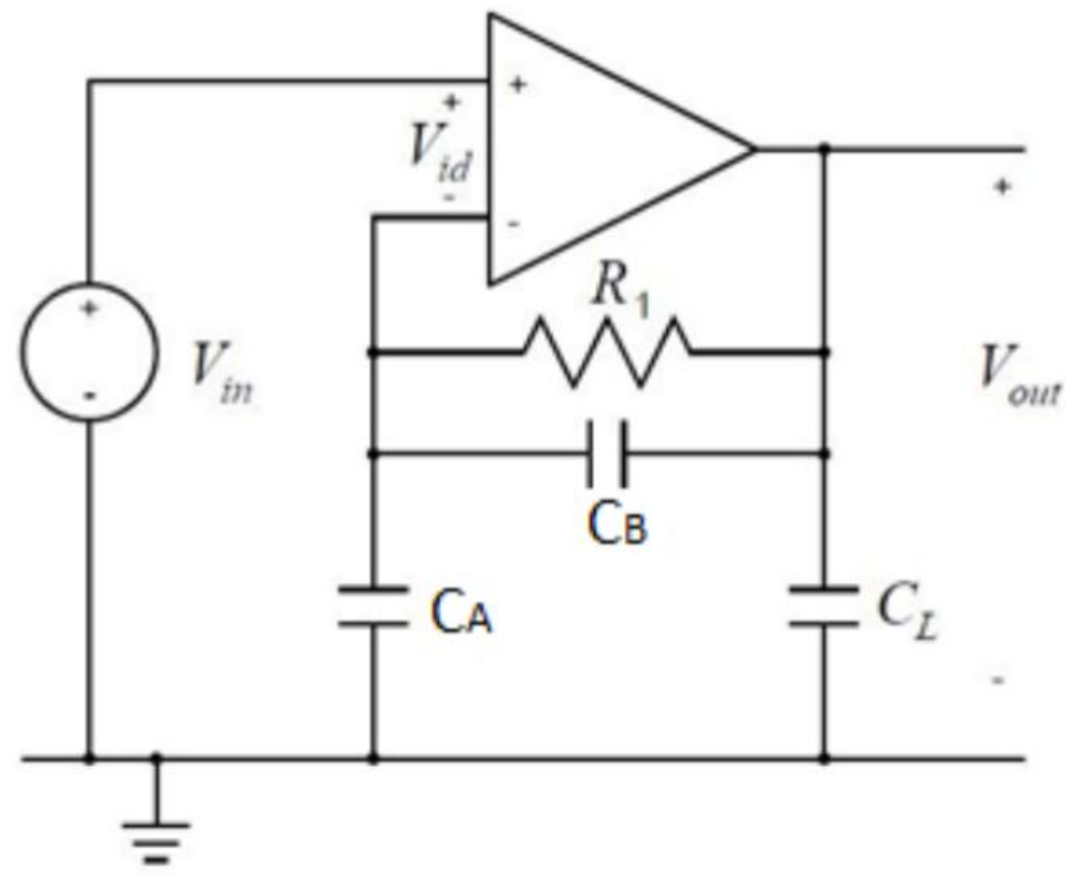


Figure 5: Schematic of a closed-loop two-stage OpAmp in a non-inverting configuration.

Design

To design the two-stage OpAmp, methods explained in textbooks by Tony Chan Carusone and Erik Bruun was used to create a preliminary design that was simulated in LTspice.

Afterward, the design was recreated inside Cadence using models that follow the 180 nm process and hand-tuned to fit the requirements. The circuit is shown to the left.

Simulation

To verify that our design meets all requirements, simulations in Cadence were performed and are shown to the right. The following results were obtained from the simulations:

Midband gain	Slew Rate	Phase margin	CL bandwidth
2.07 V/V	33.2 V/µs	93°	20.1 MHz

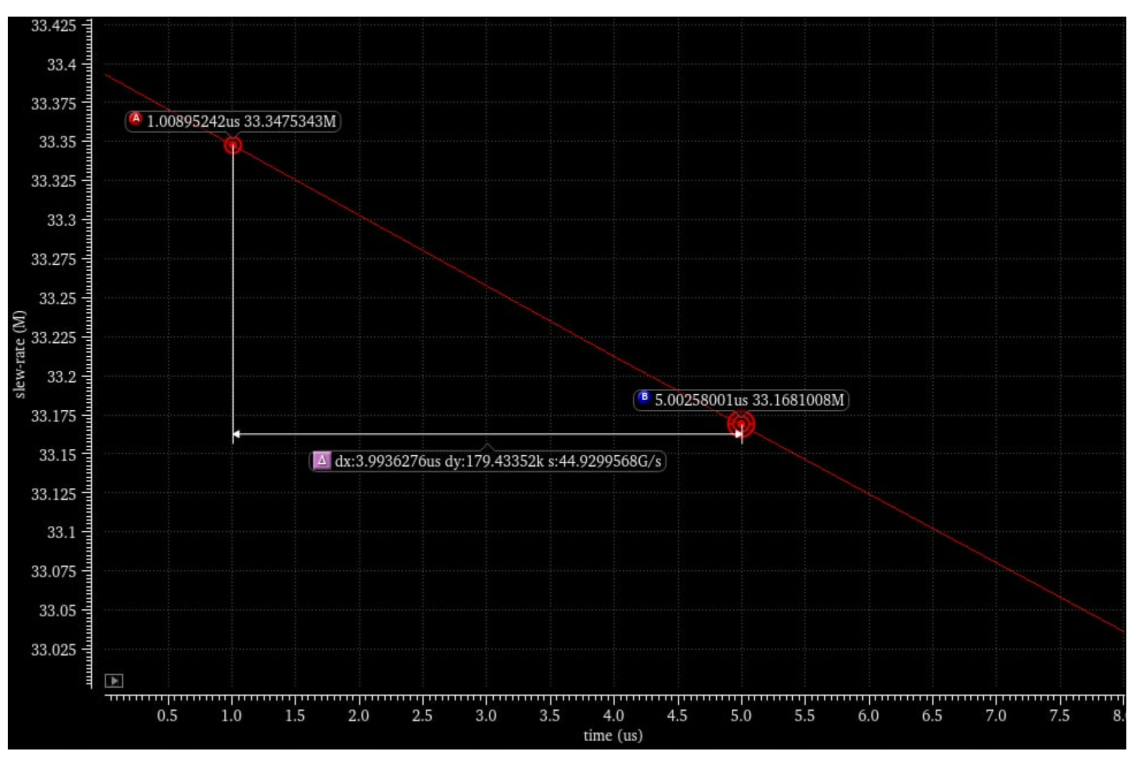


Figure 7: Shows the slew rate over time given a step function input

Open loop

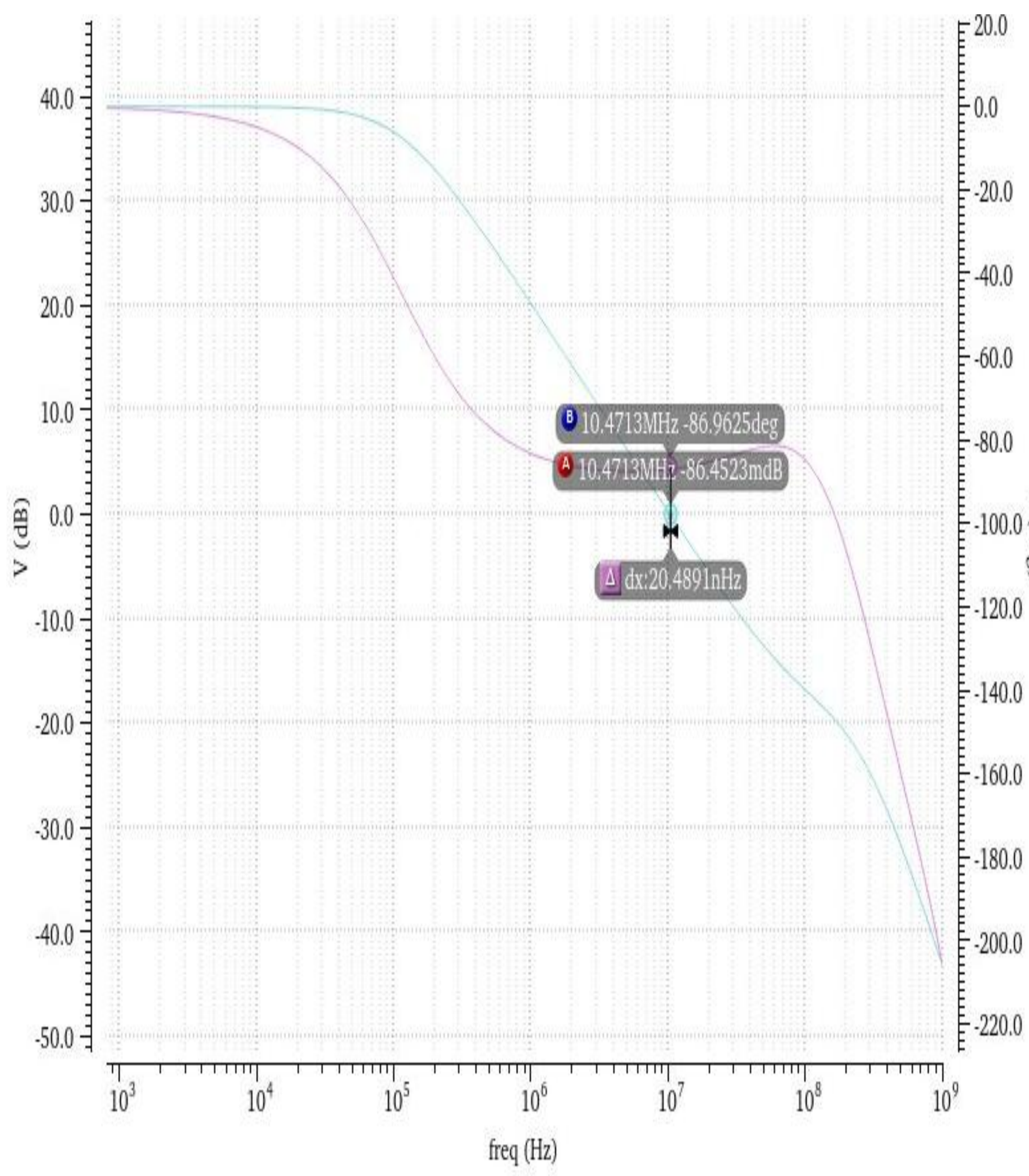


Figure 8: Bode plot of the open loop gain and phase response.

Closed loop

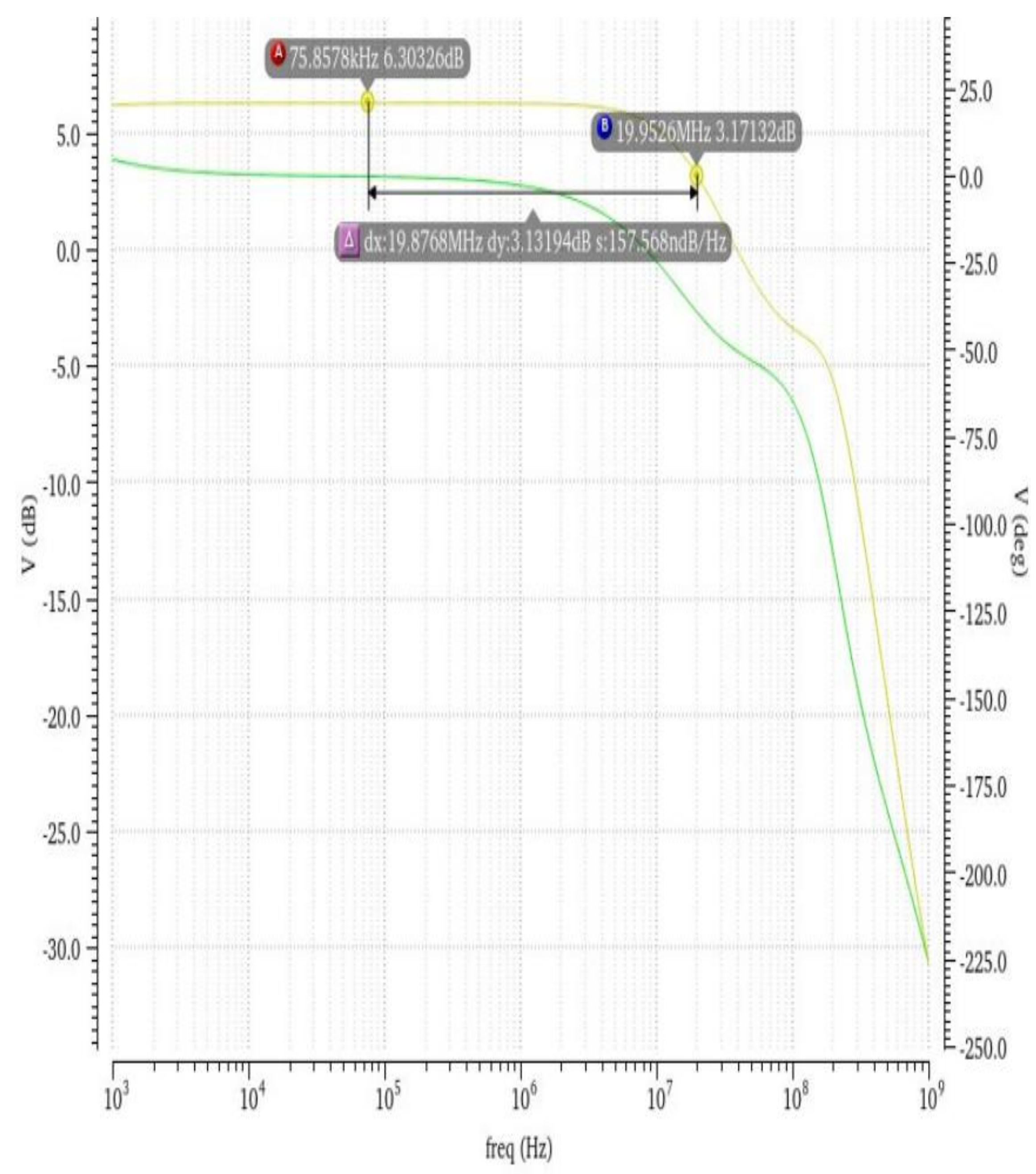


Figure 9: Bode plot of the closed loop gain and phase response.

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